THE DIMENSION OF THE SUPPORT OF A FLAT FINITE TYPE SHEAF CANNOT JUMP ABOVE ITS GENERIC VALUE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5255-5262 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 5264-5273, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Going-down for flat modules over primes (Algebra), and the lemma relating specializations of associated points to dimensions of supports (same chapter); cited inputs with the specialization argument written in the retained span.

Source reference. Original lemma 080T: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5273–5273. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GSI: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5167–5177. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $\mathcal{F}$ be a finite type, quasi-coherent $\mathcal{O}_X$-module flat over S. Assume S is irreducible with generic point η. If $\dim(Supp(\mathcal{F}_\eta)) \leq r$ then for all $s \in S$ we have $\dim(Supp(\mathcal{F}_s)) \leq r$.*

Proof. Let $x \in \mathrm{Supp}(\mathcal{F}_s)$ be a generic point of an irreducible component of $\mathrm{Supp}(\mathcal{F}_s)$. By Algebra, Lemma 080T we can find a specialization $y \rightsquigarrow x$ in $\mathrm{Supp}(\mathcal{F})$ with $f(y) = \eta$. Of course we may assume y is a generic point of an irreducible component of $\mathrm{Supp}(\mathcal{F}_\eta)$. We conclude from Lemma 0GSI that the dimension of $\overline{\{x\}}$ is at most r. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.